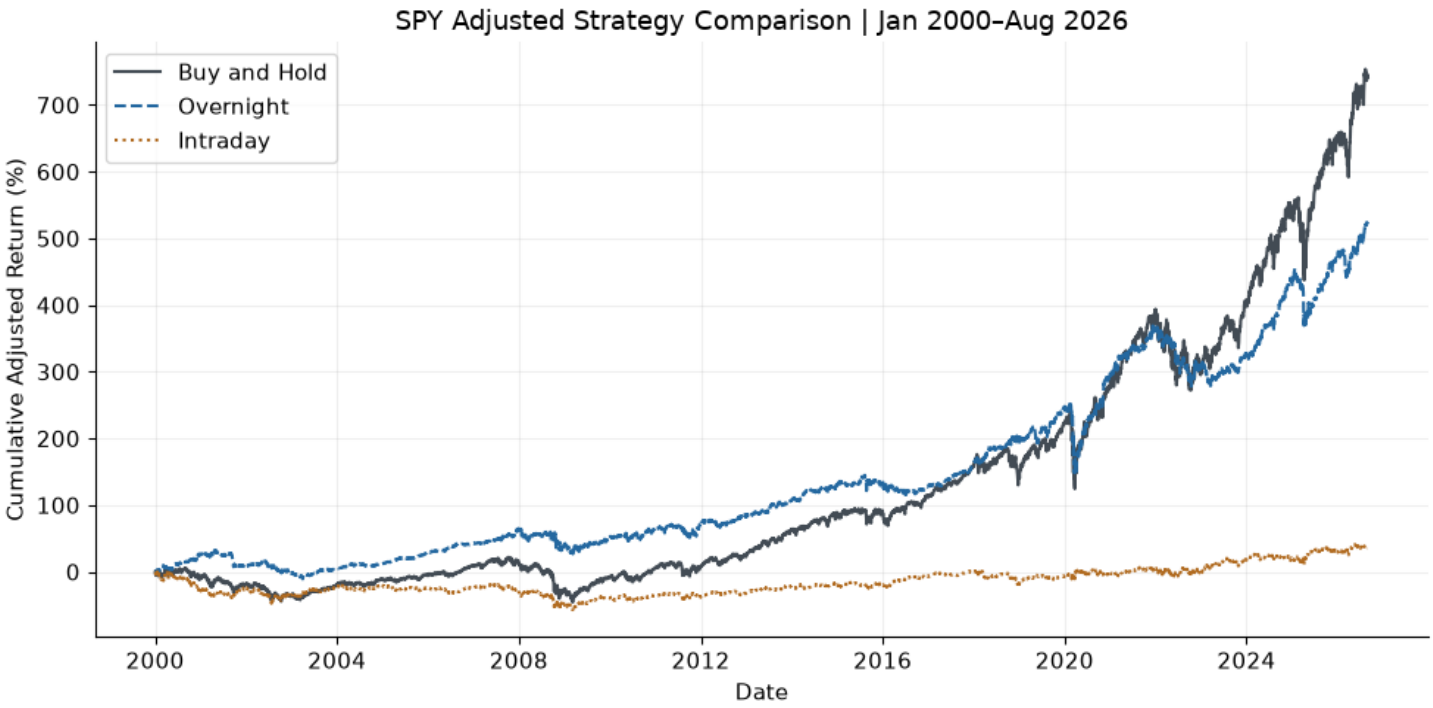
PYTHON PROJECT / RESEARCH SUMMARY

Overnight Market Anomaly

When do equity returns occur, and what risks come with them?

SPY focus: Jan 2000-Aug 2026 | 6,704 daily return observations
Adjusted prices throughout; also compares QQQ, IWM, and DIA.



SPY PERFORMANCE SNAPSHOT

Measure	Overnight-only	Buy-and-hold
Final value of $1	$6.23	$8.42
Annualized Sharpe ratio	0.67	0.51
Maximum drawdown	32.79%	55.19%

Return timing. Overnight returns contributed 85.91% of SPY's cumulative log adjusted return. Buy-and-hold still finished with more wealth.

Statistical uncertainty. The paired test did not establish a difference between mean overnight and intraday returns at the 5% level (p = 0.1577).

Tail sensitivity. Removing the top 1% of overnight observations reduced growth from 6.23x to 0.94x. This hindsight exercise is not a trading rule.

Basis: adjusted SPY prices, a dividend-inclusive return proxy; baseline excludes trading costs. Sharpe assumes 252 trading days and a zero risk-free rate. Risk comparisons are descriptive; the test does not account for dependence across days.

Source: Yahoo Finance via yfinance | Tools: pandas, NumPy, SciPy, Matplotlib

Full notebook, data, methods, and limitations: view the GitHub repository